$a_1 = \boldsymbol{p}_1 \cdot \bar{\boldsymbol{p}} = 1$

$a_2 = \boldsymbol{p}_2 \cdot \bar{\boldsymbol{p}} = 0$

$a_3 = \boldsymbol{p}_3 \cdot \bar{\boldsymbol{p}} = -1$

Collective heading direction of the school ($\bar{\boldsymbol{p}}$)

Heading direction of fish i ($\boldsymbol{p}_i$)